

Lec 51 Classifier free Diffusion Models

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Transcript

[00:00] Hello everyone. Welcome back to the lecture series on fundamentals of generative AI and large language models. In my previous session, I have discussed about classifier guided diffusion models. In in that session, we have seen what are the limitations of classifier guided diffusion model. In today's session, I will discuss about another architecture, which is classifier free diffusion guidance, okay? So, we will look into the contents for today's discussion.

[00:32] So, it starts with motivation for classifier free diffusion guidance, which we call as CFG. Concept of classifier free diffusion guidance. Joint training of the neural network. Mathematical formulation of CFG. Sampling process. And comparison between classifier guided diffusion and classifier free diffusion guidance, okay? So, we will start with the motivation for classifier free diffusion guidance. So, as we have seen in my previous session, the main limitations of classifier guided diffusion model is it requires training of an additional classifier.

[01:13] So, basically, what we are doing here is we are in the process of guiding the model for a particular type of you know, particular class of image generation, we are adding an external classifier. That increases the complexity of the model. So,

it requires training of an additional classifier. That to specifically on noisy images, because we require that gradient for tuning my generated samples towards that particular class.

[01:47] And also, we have seen that off-the-shelf classifiers like ResNet, which is trained in ImageNet data set, they cannot be used. Because these pre-trained models or you know, off-the-shelf classifiers, they are trained in clean images. So, they will not work well with noisy images, okay? So, this two things, the um a classifier as well as the diffusion model, unconditional diffusion model, this makes the entire process very complex, okay? And it requires more time also during because there is training of the classifier as well as you require to train the DDPM, okay?

[02:34] Now, we will look into how this external classifier or without using an explicit classifier, can we get the similar kind of guidance? And that is the concept of classifier-free diffusion guidance, okay? So, what is that one? We will have a look there. » [snorts] » The concept was introduced in the paper classifier-free diffusion guidance by Jonathan Ho and his team from Google Research Brain Team. And this is the reference for the paper. It was presented at NeurIPS workshop in 2022.

[03:13] And we will see what is the core idea for this paper or for this work. So, in this uh concept, what the core idea is, it is to do a conditional generation without training an external classifier. That is the first thing. We want to get rid of the external classifier. In classifier-free guidance, the diffusion model is trained to perform both conditional and unconditional generations simultaneously. So, in order to get rid of the classifier, which is the first and foremost requirement, What we do is a single diffusion model is trained to perform both conditional and unconditional generation simultaneously, okay?

[04:03] The diffusion model then learns both conditional as well as unconditional denoising, allowing guidance to be obtained without a separate classifier. So, the main idea is first of all, we are removing the classifier entirely from this picture. And then we are using the same network to train for conditional as well as unconditional generation. And then what we do is we use this both conditional and unconditional denoising, and the difference between them we use for steering for a specific class, okay?

[04:40] Now, there is something called as the joint training of the neural network. What is this joint training? As I was telling, the same neural network will be trained both for your conditional as well as unconditional generation. So, what exactly we do here is we have the neural network, then we have class labels, which we supply for conditional training. Okay? And when we give class labels for conditional training, what we expect is we get a conditional score, okay? So, this conditional score is dependent on the class label. So, your network is taking the condition and giving you some conditional score.

[05:17] Together with that one, you want an unconditional score also. For that, what you are doing? Class labels are dropped with a fixed probability, okay? So, you are randomly dropping the class label, maybe 10% and 20 or 20% of the data, and these class labels are replaced by a null token, okay? So, this is replaced by a null token, so that that same network becomes unconditional. Okay? So, we have one neural network only where class labels are you know, supplied for conditional score generation and those class labels are randomly dropped out from some portion of the data and they are replaced by a null token which is which gives you the unconditional score. Okay?

[06:12] Then train a conditioned diffusion model which depends on x_{t+1} with condition dropout epsilon theta x_t phi. So, what exactly is happening is so, when you are doing this conditional score thing, okay? When you are working with a conditional score, you get the uh noise estimate which depends on x_t which is your noisy sample. T is the time step, Y is your class label. Okay? And then you drop out that condition from some part of the data and you get some noise estimate which is dependent on x_t , t and ϕ .

[06:57] What is ϕ ? ϕ basically is the null token. So, what you are doing is Y is removed to make the model unconditional. So, actually Y is removed and replaced. Okay? So, it is replaced by ϕ . So, you are removing the Y and then Y is replaced by ϕ . What is ϕ ? ϕ is your null token and it represents the absence of conditioning information. Okay? So, this is how one neural network itself can be trained for conditional score generation as well as unconditional score generation and we get two different noise predictions, and that we use for guiding the generation towards a specific class. Okay?

[07:50] So, that is the main idea. Now, we will see into the mathematical formulation. So, if we take $P(Y \text{ given } X_T)$, which is your probability of getting Y class given you have a noisy image X_T , that we get from the classifier, that we will try to write in terms of unconditional and conditional unconditional DDPM. We will see how to write that one. So, this $P(Y \text{ given } X_T)$ in my classifier guided diffusion, we have seen this is the probability that is coming from the classifier. Okay?

[08:30] Because when you take the gradient, this term was there. Now, if I write this one, this is the probability of getting class label Y given the noisy image X_T . Now, using base theorem, we can write $P(Y \text{ given } X_T)$ equal to $P(X_T \text{ given } Y)$ into $P(Y)$ by $P(X_T)$. Okay? $P(X_T)$. Now, taking log- logarithm in both sides, what do we get? How can we write $\log P(Y \text{ given } X_T)$? This is my first term with log. $P(X_T \text{ given } Y)$, the first top term, plus log of $P(Y)$. This is this term, and log of $P(X_T)$.

[09:14] Okay? So, uh sorry, this is minus log of $P(X_T)$. Okay? Because it is in the denomina- uh nume- uh denominator. Okay? Now, taking gradient with respect to X_T on both sides, same way as we did in the classifier guided diffusion models. So,

what we do here is we would like to take the for all these terms. And then write the next step of the equation. » [snorts] » Now, p_y is constant. We know there is no dependency of p_y on x_t . So, when we take the gradient with respect to x_t , this p_y term becomes zero. Okay? The gradient with respect to x_t is zero. So, then we can write it like gradient with respect to $x_t \log p_y$ given x_t can be written as gradient with respect to $x_t p \log p$ given y plus

[10:13] gradient of uh you know with respect to $x_t \log p$ of x_t . So, what we are doing? This we were supposed to get from the classifier. In classifier-guided diffusion, we have seen this was the gradient that was coming from the classifier. So, that can be written as a summation of two terms. One is the conditional term and another one is the unconditional term. So, if you relate with my previous slide image that same unit or same neural network, I am once training it for conditional score generation, once for unconditional score generation. Using those two, I can write the classifier guidance term. So, this was this was my classifier.

[11:05] » [snorts] » Uh you know, um classifier gradient term. So, this classifier gradient term can be written as a combination of conditional term and unconditional term that is coming from the single neural network. Okay? So, that is the main idea and this is captured so beautifully with the mathematical formulation. So, this classifier free guidance approximates this classifier gradient using the diffusion model itself. So, here if you see, there is no external classifier that I have used. I am what I'm replacing is whatever I was getting from the external classifier, this term.

[11:50] This term was the one which I was getting from the external classifier. That I'm replacing or writing in terms of conditional and unconditional term that I can get from the same neural network, okay? Now, coming to the noise prediction, okay? The noise prediction equation is given as $\epsilon_t \hat{\epsilon}_t$, this is your new noise predicted. $\hat{\epsilon}_t$ which depends on x_{t-1} and y , okay? is equal to $1 + s \epsilon_t$ not x_{t-1} minus $s \epsilon_t$ not x_{t-1} ϕ s is the classifier guidance. I'm not going into the derivation part of this math of this equation. What I want to tell here is it's a combination of the noise that you get from your conditional part and what you get from your unconditional part. If you remember, the

[12:54] unconditional part we got by replacing the y with ϕ . Okay? So, this is coming from your this noise is coming from your conditional part conditional, you know, training and this term is coming from your unconditional training, okay? So, that is the new noise that is getting predicted. And this new noise, as we all know, will help in steering the generation towards the specific class. So, what we are doing is we are taking a combination of this noise without condition and the noise with condition and we are predicting the new noise.

[13:38] Now, sampling from the classifier-free guidance model, how it happens. Start the sampling Start by sampling data with conditioning from the data set.

Randomly discard the conditioning to train the unconditional version of the same network. That has been done. » [snorts] » Train the same network using both conditional and unconditional examples. Okay? So, that is the training part and then what happens? This teaches the diffusion model to be able to guide the model towards generating samples from a class without explicitly needing a classifier. So, So, we start the sampling data from conditioning from the data set with conditioning from the data set, randomly discard and train it for unconditional

[14:24] thing, and train the same network using both conditional and unconditional samples, and then use that noise predicted to help in the uh you know, uh generating images that are aligned to a particular class. Okay? This is an image that I have taken from the original paper. And what we have seen is in the left side images, this side images, » [snorts] » they are they are the non-guided samples, whereas the right side images, they are the classifier-free guidance with with classifier-free uh guided model with s equal to 3. So, if you see the left side ones, they are not very convincing. The results are not very convincing if you see the left side images. While the right side images are pretty convincing that they are from a

[15:24] particular class, okay? So, it shows that classifier-free guided samples, even though there is a little bit of saturation of color, but it gives very precisely that class information is there in that image, okay? So, I have taken this one from the original paper of Ho et al. classifier-free diffusion guidance, which was presented at NeurIPS workshop in 2022. And this shows classifier-free guided samples, and the and it is compared with non-guided samples. So, we are here we are doing it without explicitly using the classifier.

[16:10] Now, a small comparison table between the classifier guided and classifier-free guidance. So, separate classifier is required in classifier guided, yes. That is no, we don't require and that is the major advantage for classifier-free guidance. We don't want to make the pipeline model pipeline more complex. We want to keep it as as simple as possible. So, we have removed the classifier from them. Noisy classifier training is required in classifier guided guided uh model. It is not required in classifier-free guidance.

[16:49] Guidance score is coming from the classifier gradient in case of classifier guided models, where whereas it is the difference between the conditional and unconditional. That difference, conditional minus unconditional, that difference helps to get the guidance score, okay? And the sampling cost in this, it is unit plus classifier, both needs to be training trained during the you know, when you want to generate samples. While in classifier-free guidance, you need two unit passes.

[17:25] Why two unit passes are required? We know unit architecture is used for denoising in any diffusion model. Now, in this case, we don't have a classifier, but we have two tasks in hand. One is to train the unit for conditional, uh, you know, with

conditional samples. Our samples with a particular conditional information given with them. And also, you have to train the same unit with unconditional samples, okay? That means for each and every image, you need to train it twice.

[18:03] Once for condition, another one is the unconditional one. So, that is the comparison between classifier-guided and classifier-free guidance diffusion models. With this one, I come to the end of today's session. And before I conclude, let us summarize. So, what we have learned today? So, we know what are the limitations of classifier-guided diffusion models. There is an extra classifier that needs to be trained. We need to, we want to get rid of that one to make the pipeline more simple, okay?

[18:33] For that one, the core concept of classifier-free guidance came into picture. So, what is that one? It is the core idea is to use one network itself, use one unit itself for training both for conditional as well as unconditional samples. And once you do that one, once once you train the unit for the conditional one, you get one noise term. Once you get train it for um, unconditional, uh, samples, you get another noise prediction, which is for the unconditional samples. So, you get two noise predictions. And the difference between these two scores will give you will guide you will guide the model for generating towards the conditioned class. Okay. So, that is the main idea and there are some

[19:25] mathematical steps that have been done and it captures how the input that I required from the classifier side, particularly the gradient of $u_h p$ of $u_h y$ given x that we can write in terms of two terms, that is that can come from the u_h unit architecture one network itself, the conditional and the unconditional. Using that one, we have seen how beautifully the classifier explicit presence of the classifier can be avoided and we can go for a classifier-free guidance diffusion model and we have seen how the output looks like with no guidance at all and with a classifier-free guidance. So, we have seen that picture from the um original paper. So, with this one, I come to the conclusion of today's

[20:18] session. I hope this uh the session was fruitful for you all. And in my next session, I will be speaking about stable diffusion model. Till that time, happy learning and thank you all. »